

Polchinski Renormalization Group, dynamical Γ -calculus, and transportation of measures.

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- Introduction with the continuum ϕ^4 model
- Renormalization
- Bauerschmidt-Bodineau approach to RG
- Dynamical Γ -calculus
- Transportation of measures

Introduction : the continuum ϕ^4 model

A model example : the continuous ϕ^4 model

We want $\phi : \mathbb{R}^d \rightarrow \mathbb{R}$, to be a random function with law

$$\mathcal{D}\phi \exp \left\{ - \int_{\mathbb{R}^d} dx \left(\frac{K}{2} |\nabla \phi|^2 + g(\phi^2 - 1)^2 \right) \right\}.$$

A model example : the continuous ϕ^4 model

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—> $\mathcal{D}\phi$ is the Feynman path-integral on $\{\mathbb{R}^d \rightarrow \mathbb{R}\}$,
i.e. the Lebesgue measure in infinite dimension : **ill-defined !**

A model example : the continuous ϕ^4 model

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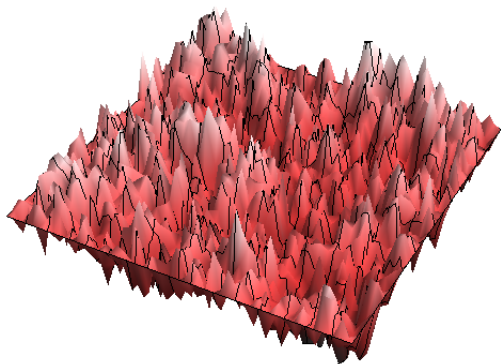
$\longrightarrow \mathcal{D}\phi \exp \left\{ - \frac{K}{2} \int_{\mathbb{R}^d} dx |\nabla \phi|^2 \right\}$ Gaussian Free Field on $\mathbb{R}^d$,

i.e. the Brownian motion with a d -dimensional time

well-defined as a distribution for $d > 1$

and as a continuous function for $d = 1$

The continuous ϕ^4 model



A realization of the GFF is a H^{-1} -fluctuation of a constant function, which is a minimizer of GFF's energy (image : Wikipedia).

A model example : the continuous ϕ^4 model

We want $\phi : \mathbb{R}^d \rightarrow \mathbb{R}$, to be a random function with law

$$\mathcal{D}\phi \exp \left\{ - \int_{\mathbb{R}^d} dx \left(\frac{K}{2} |\nabla \phi|^2 + g(\phi^2 - 1)^2 \right) \right\}.$$

—> $H(\phi) := -g \int_{\mathbb{R}^d} dx (g(\phi^2 - 1)^2)$ is a self-interaction

—> g is the associated quartic interaction constant

—> when $g \rightarrow \infty$, $\phi = \pm 1$, gives the Ising model

A model example : the continuous ϕ^4 model

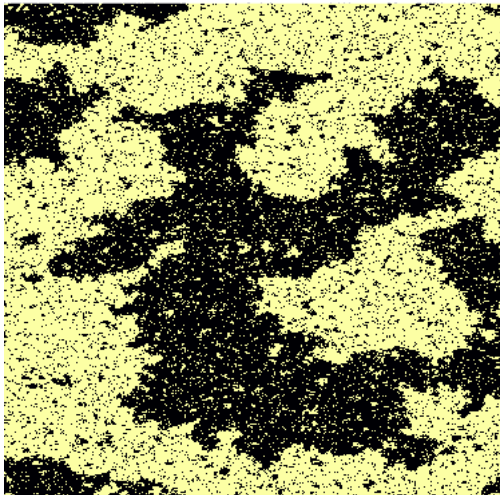


Image : Physics Simulations on GitHub

Renormalization

Renormalization

Problems

No Lebesgue measure in infinite dimension.

→ Coarse-graining : make ϕ constant below a scale r .
No fluctuations above an energy scale $\Lambda \sim 1/r$.

$$\bar{\phi}(x) = \frac{1}{|B_r(x)|} \int_{B_r(x)} \phi(\zeta) d\zeta$$

→ Rescale in the new variable $x' = x/r$, and normalize with contrast ξ

$$\tilde{\phi}(x') = \frac{1}{\xi} \bar{\phi}(rx)$$

→ In Fourier space : high frequencies cutoff.

Kadanoff block spin picture, 1966

Ising model on lattice

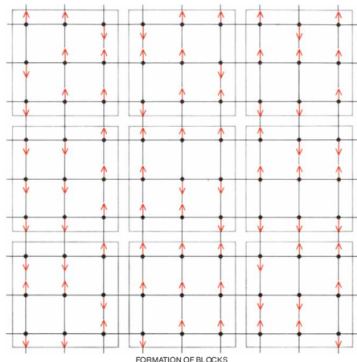


Image from Wilson, K. G. (1979). Problems in physics with many scales of length. Scientific American.

Kadanoff block spin picture, 1966

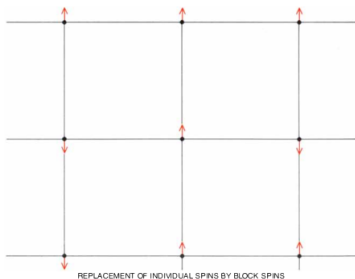


Image from Wilson, K. G. (1979). Problems in physics with many scales of length. Scientific American.

Kadanoff block spin picture, 1966

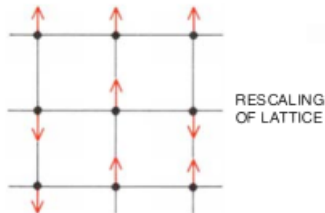


Image from Wilson, K. G. (1979). Problems in physics with many scales of length. Scientific American.

Exact Renormalization Group, Wilson, 1971

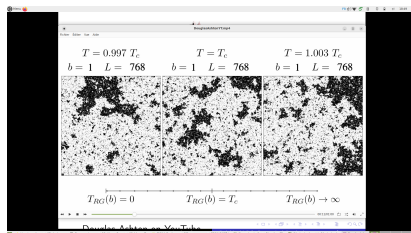
Renormalization as a continuous flow in parameter space.

—> It introduces interactions at all orders. The Hamiltonian

$$\int_{R^d} dx \left[\frac{t}{2} \phi(x)^2 + u \phi(x)^4 + \dots + \frac{K}{2} |\nabla \phi(x)|^2 + \frac{L}{2} |\nabla^2 \phi(x)|_{HS}^2 + \dots \right]$$

$$\text{Infinite system of differential equations} \quad \left\{ \begin{array}{l} dt/dr = \dots \\ du/dr = \dots \\ \vdots \\ dK/dr = \dots \\ dL/dr = \dots \\ \vdots \end{array} \right.$$

Illustration ERG on the 2d-Ising model

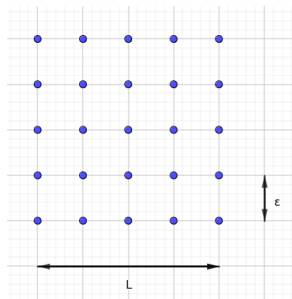


Douglas Ashton on YouTube

The constructive approach

$$\mathbb{R}^d \longleftrightarrow \Lambda = L\mathbb{T}^d \cap \varepsilon\mathbb{Z}^d$$

functions on $\Lambda \longleftrightarrow$ vectors in $\mathbb{R}^{|\Lambda|}$



Philosophy

Properties that are independent of ε will remain true for the continuum model.

The constructive approach

For the ϕ^4 model :

- for all $d \geq 4$, $\phi_d^4 = \text{GFF}$
- for $d = 4$, $\phi_4^4 = \text{GFF}$ [Aizenman and Duminil-Copin, 21]
- for $d = 2$ or 3 , $\phi_d^4 \neq \text{GFF}$
- for $d = 1$, ϕ_1^4 function, $\text{GFF} = \text{Brownian}$ and $\phi_1^4 \neq \text{GFF}$.

*The Polchinski Renormalization Group
by Bauerschmidt and Bodineau*

We consider a random function on the lattice, with law

$$\nu_0(d\phi) = \frac{1}{Z} \gamma_{C_\infty}(d\phi) \exp(-V_0(\phi)), \quad \phi \in \mathbb{R}^d$$

- $d = |\Lambda|$
- C_∞ is the discrete Laplacian on Λ
- γ_{C_∞} is the discrete GFF on $\Lambda \Rightarrow$ Gaussian on $\mathbb{R}^p \subset \mathbb{R}^d$
- $V_0(\phi) = \sum_{x \in \Lambda} V(\phi_x)$ are the self-interactions

Polchinski RG by Bauerschmidt and Bodineau

$$\nu_0(d\phi) = \frac{1}{Z} \gamma_{C_\infty}(d\phi) \exp(-V_0(\phi)), \quad \phi \in \mathbb{R}^d$$

We decompose the covariance

$$C_\infty = \int_0^\infty \dot{C}_t dt, \quad \dot{C}_t \geq 0, \quad C_t : 0 \nearrow C_\infty.$$

Two main decompositions :

- heat kernel decomposition $\dot{C}_t = e^{-tC_\infty^{-1}}$
→ block size increases as the discrete Brownian spreads
- Pauli-Villars decomposition $C_t = (C_\infty^{-1} + 1/t)^{-1}$
→ fictitious particles of mass $t \rightarrow \infty$

Polchinski RG by Bauerschmidt and Bodineau

We can compute

$$\begin{aligned}\mathbb{E}_{\nu_0} [F(\phi)] &\simeq \mathbb{E}_{C_\infty} [e^{-V_0(\phi)} F(\phi)] \\ &= \mathbb{E}_{C_\infty - C_t} [\mathbb{E}_{C_t} [e^{-V_0(\varphi+\zeta)} F(\varphi + \zeta)]] \\ &= \mathbb{E}_{C_\infty - C_t} \left[Z_t(\varphi) \frac{1}{Z_t(\varphi)} \mathbb{E}_{C_t} [e^{-V_0(\varphi+\zeta)} F(\varphi + \zeta)] \right]\end{aligned}$$

where C_t acts on ζ and $C_\infty - C_t$ acts on φ . Then we set

$$V_t(\varphi) := -\log Z_t(\varphi) = -\log \mathbb{E}_{C_t} (e^{-V_0(\varphi+\zeta)})$$

$$\nu_t(d\varphi) \simeq \gamma_{C_\infty - C_t}(d\varphi) e^{-V_t(\varphi)}$$

$$\mathbb{E}_{\mu_t^\varphi} [F(\zeta)] = \frac{1}{Z_t(\varphi)} \mathbb{E}_{C_t} [e^{-V_0(\varphi+\zeta)} F(\varphi + \zeta)]$$

Polchinski RG by Bauerschmidt and Bodineau

decomposition of ν_0

$$\mathbb{E}_{\nu_0} [F(\phi)] = \mathbb{E}_{\nu_t} [\mathbb{E}_{\mu_t^\varphi} [F(\zeta)]]$$

Renormalized potential

$$V_t(\varphi) := -\log \mathbb{E}_{C_t} (e^{-V_0(\varphi+\zeta)})$$

Polchinski flow

$$\nu_t(d\varphi) \simeq \gamma_{C_\infty - C_t}(d\varphi) e^{-V_t(\varphi)}$$

Fluctuation measure

$$\mathbb{E}_{\mu_t^\varphi} [F(\zeta)] = \frac{1}{Z_t(\varphi)} \mathbb{E}_{C_t} [e^{-V_0(\varphi+\zeta)} F(\varphi + \zeta)]$$

Renormalized potential

$$V_t(\varphi) := -\log \mathbb{E}_{C_t} (e^{-V_0(\varphi+\zeta)})$$

- It satisfies the Hamilton-Jacobi equation

$$2\partial_t V_t = \Delta_{\dot{C}_t} V_t - |\nabla V_t|_{\dot{C}_t}^2$$

which is particular case of the Polchinski equation.

- V_∞ is equivalent to the log-Laplace transform of ν_0 .

Polchinski flow

$$\nu_t(d\varphi) \simeq \gamma_{C_\infty - C_t}(d\varphi) e^{-V_t(\varphi)}$$

- ν_0 is the original measure,
- $\nu_\infty = \delta_0$ is a Dirac distribution
- can be realized as a Follmer process
(process conditioned on the future)

Fluctuation measure

$$\mathbb{E}_{\mu_t^\varphi} [F(\zeta)] = \frac{1}{Z_t(\varphi)} \mathbb{E}_{C_t} [e^{-V_0(\varphi+\zeta)} F(\varphi + \zeta)]$$

- $\mu_0^\varphi = \delta_\varphi$, with $\varphi \sim \nu_0$
- $\mu_\infty^\varphi = \nu_0$ up to the external field $C_\infty^{-1}\varphi$
- up to time inversion, and good choice of $\varphi = \varphi_t$, this is Eldan's stochastic localization

What is the logarithmic Sobolev constant ?

- > Let μ be a probability distribution
- > A Markov process X_t with equilibrium distribution μ

LSI constant ρ = the exponential relaxation time in entropy :

$$\text{Ent}_\mu [\mathcal{L}(X_t)] \leq e^{-\frac{4t}{\rho}} \text{Ent}_\mu [\mathcal{L}(X_0)]$$

(Convergence in $L^1 \log L^1$)

Bauerschmidt-Bodineau multiscale criterion

Thm (Bauerschmidt-Bodineau 20)

Let $\nu_0(d\phi) = \frac{1}{Z} \gamma_{C_\infty}(d\phi) \exp(-V_0(\phi))$ be a probability,
 C_t a covariance decomposition, and
 $V_t(\varphi) = -\log \mathbb{E}_{C_t}(e^{-V_0(\varphi+\zeta)})$ the renormalized potential.
If for all $t \geq 0$,

$$\dot{C}_t \nabla^2 V_t \dot{C}_t - \frac{1}{2} \ddot{C}_t \geq \dot{\lambda}_t \dot{C}_t$$

for some $\dot{\lambda}_t \in \mathbb{R}$, then the LSI constant of ν_0 satisfies

$$\rho \leq 2|\dot{C}_0| \int_0^\infty e^{-2\lambda_s} ds$$

$$\text{If } \dot{C}_t \nabla^2 V_t \dot{C}_t - \frac{1}{2} \ddot{C}_t \geq \dot{\lambda}_t \dot{C}_t$$

$$\text{then } \rho \leq 2 |\dot{C}_0| \int_0^\infty e^{-2\lambda_s} ds$$

- If V_0 is convex, then $\forall t \geq 0$, V_t remains convex (Prekopa-Leindler or maximum principle)
- Take the heat kernel decomposition $\dot{C}_t = e^{-tA}$, $A = C_\infty^{-1}$, then the criterion works for $\dot{\lambda}_t = K_{\min}(A)/2$ and gives

$$\rho \leq 2 \int_0^\infty e^{-K_{\min}(A)s} ds = \frac{2}{K_{\min}(A)}$$

which recovers the usual Bakry-Émery criterion.

Application to the ϕ_d^4 models, $d = 2$ or 3 .

Thm (Bauerschmidt-Dagallier 22)

For all temperature $T > T_c$, the ϕ_d^4 measure ($d = 2, 3$) on the lattice $\Lambda_{\varepsilon, L}$ has a log-Sobolev constant which is bounded uniformly in ε .

Proof involves precise correlation inequalities for the Ising models with external fields, and bounds on the susceptibility of the ϕ_d^4 ($d=2,3$) models.

Evolution of spectral quantities under inhomogeneous flow

Evolution of quantities under inhomogeneous flow

→ deterministic evolution :

$$x_t = x_0 + \int_0^t \dot{x}_s ds$$

→ stochastic homogeneous Markov :

$$\forall f, \quad f(X_t) = f(X_0) + \int_0^t \mathcal{L}f(X_s) ds + \text{martingale}$$

with $\mathcal{L}$ a second-order differential operator

Example :

- Δ for the Brownian motion,
- $\mathcal{L} = \Delta - x \cdot \nabla$ for the Ornstein-Uhlenbeck process.

Evolution of quantities under inhomogeneous flow

→ stochastic homogeneous Markov :

$$\forall f, \quad f(X_t) = f(X_0) + \int_0^t \mathcal{L}f(X_s) ds + \text{martingale}$$

→ stochastic inhomogeneous Markov :

$$\forall f, \quad f(X_t) = f(X_0) + \int_0^t \mathcal{L}_s f(X_s) ds + \text{martingale}$$

with $\mathcal{L}_s$ a family of second-order differential operators

Example : non-homogeneous Ornstein-Uhlenbeck

$$\mathcal{L}_s = \Delta - A_s x \cdot \nabla$$

Evolution of quantities under inhomogeneous flow

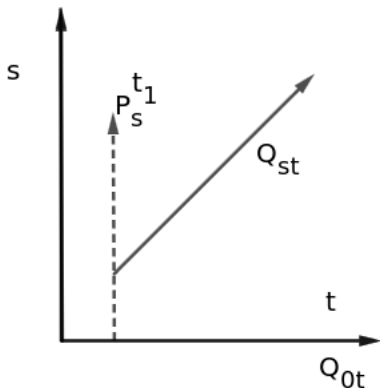
- X_t be an inhomogeneous Markov process
- $\mathcal{L}_t$ its generator
- ν_t law of X_t
- $Q_{st} : f \mapsto (x \mapsto \mathbb{E}[f(X_t)|X_s = x])$ the semigroup
- $2\Gamma_t(f, g) = \mathcal{L}(fg) - f\mathcal{L}g - g\mathcal{L}f$ the carré du champ
 - > measures the extent to which $\mathcal{L}_t$ is first order
 - > in some sense, in average $\Gamma_t(f, g) = f\mathcal{L}_t g$
- $2\Gamma_{2,t}(f, g) = \mathcal{L}(\Gamma_t(f, g)) - \Gamma_t(f, \mathcal{L}g) - \Gamma(g, \mathcal{L}f)$
 - > measures the extent to which Γ_t and $\mathcal{L}_t$ commutes

Evolution of quantities under inhomogeneous flow

Now, for a fixed t , we consider

- $(Y_t)_s$ be an homogeneous Markov process
- with equilibrium distribution ν_t
- Λ_t its generator
- $(P^t_s)_{s \geq 0}$ its semigroup
- Γ_t^Λ its carré du champ
- $\Gamma_{2,t}^\Lambda$ its Γ_2 operator

Evolution of quantities under inhomogeneous flow



Evolution with two-dimensional time.

What is the Poincaré constant ?

- > Let μ be a probability distribution
- > A Markov process X_t with equilibrium distribution μ

Poincaré constant C_P = the exponential relaxation time in L^2 :
for all functions f ,

$$\mathrm{Var}_\mu [f(X_t)] \leq e^{-\frac{t}{C_P}} \mathrm{Var}_\mu [f(X_0)]$$

Mixing time along inhomogeneous flow

Thm (S 24)

If for all $t \geq 0$,

$$\frac{d}{dt} \Gamma_t^\Lambda + (\Gamma_{2,t}^\Lambda - 2\Gamma_{2,t}) \leq \dot{k}_t \Gamma_t$$

for some $\dot{k}_t \in \mathbb{R}$, then for all $0 \leq s \leq t$,

$$C_P(\nu_s) \leq e^{\int_s^t \dot{k}_s ds} C_P(\nu_t)$$

Mixing time along inhomogeneous flow

If for all $t \geq 0$,

$$\frac{d}{dt} \Gamma_t^\Lambda + (\Gamma_{2,t}^\Lambda - 2\Gamma_{2,t}) \leq \dot{k}_t \Gamma_t$$

Then for all $0 \leq s \leq t$,

$$C_P(\nu_s) \leq e^{\int_s^t \dot{k}_s ds} C_P(\nu_t)$$

- Homogeneous case, recover Klartag-Putterman's result :
- For $\nu_t(dx) = H_t(f) dx$ with $H_t =$ heat-flow and f log-concave, gives that C_P is non-decreasing along the heat flow (Klartag-Putterman 21).
- Can be adapted to $\nu_t =$ Polchinski flow !

Sketch of proof for the inhomogeneous O.-U.

inhomogeneous O.-U

$$dX_t = -\frac{K_t}{2}X_t dt + dB_t$$

$$\mathcal{L}_t = \frac{1}{2}\Delta - \frac{K_t}{2}x \cdot \nabla$$

Idea : differentiate the Rayleigh ratio :

$$R_\phi(t) := \frac{\mathbb{E}_{\nu_t} |\nabla Q_{0t}\phi|^2}{\mathbb{E}_{\nu_t} (Q_{0t}\phi)^2}$$

to show that

$$C_P(\nu_t) \geq e^{-\int_0^t k_s ds} C_P(\nu_0)$$

and iterate by the semigroup property (start the flow at ν_t).

Sketch of proof for the inhomogeneous O.-U.

Let's compute

$$\partial_t \mathbb{E}_{\nu_t} \phi_t^2 = 2 \mathbb{E}_{\nu_t} (\phi_t \mathcal{L}_t \phi_t) + \int \phi_t^2 \partial_t \nu_t$$

Fokker-Planck equation for the inhomogeneous O.-U :

$$\partial_t \nu_t = \frac{1}{2} \nabla \cdot [\{\nabla \log \nu_t + K_t \varphi\} \nu_t]$$

so

$$\int \phi_t^2 \partial_t \nu_t = \frac{1}{2} \int \nabla(\phi_t^2) \cdot (\nabla \log \nu_t + K_t \varphi) d\nu_t$$

Sketch of proof for the inhomogeneous O.-U.

Let's compute

$$\partial_t \mathbb{E}_{\nu_t} \phi_t^2 = 2\mathbb{E}_{\nu_t} (\phi_t \mathcal{L}_t \phi_t) + \int \phi_t^2 \partial_t \nu_t$$

$$\int \phi_t^2 \partial_t \nu_t = \frac{1}{2} \int \nabla(\phi_t^2) \cdot (\nabla \log \nu_t + K_t \varphi) d\nu_t$$

but

$$\frac{1}{2} \int \nabla(\phi_t^2) \cdot K_t \varphi d\nu_t = \int \left(\frac{1}{2} \Delta - \mathcal{L}_t \right) (\phi_t^2) d\nu_t$$

therefore

$$\begin{aligned} \int \phi_t^2 \partial_t \nu_t &= \frac{1}{2} \int (\Delta + \nabla \log \nu_t \cdot \nabla) (\phi_t^2) d\nu_t - \int \mathcal{L}_t(\phi_t^2) d\nu_t \\ &= - \int \mathcal{L}_t(\phi_t^2) d\nu_t \end{aligned}$$

Sketch of proof for the inhomogeneous O.-U.

Let's compute

$$\partial_t \mathbb{E}_{\nu_t} \phi_t^2 = 2\mathbb{E}_{\nu_t} (\phi_t \mathcal{L}_t \phi_t) + \int \phi_t^2 \partial_t \nu_t$$

$$\int \phi_t^2 \partial_t \nu_t = - \int \mathcal{L}_t(\phi_t^2) d\nu_t$$

and finally

$$\partial_t \mathbb{E}_{\nu_t} \phi_t^2 = 2\mathbb{E}_{\nu_t} (\phi_t \mathcal{L}_t \phi_t) - \mathbb{E}_{\nu_t} \mathcal{L}_t(\phi_t^2) = -\mathbb{E}_{\nu_t} |\nabla \phi_t|^2.$$

Sketch of proof for the inhomogeneous O.-U.

On the other hand,

$$\partial_t \mathbb{E}_{\nu_t} |\nabla \phi_t|^2 = 2 \mathbb{E}_{\nu_t} \langle \nabla \mathcal{L}_t \phi_t, \nabla \phi_t \rangle + \int |\nabla \phi_t|^2 \partial_t \nu_t$$

and

$$\begin{aligned} B &= \int |\nabla \phi_t|^2 \partial_t \nu_t \\ &= \frac{1}{2} \int |\nabla \phi_t|^2 \nabla \cdot [\{\nabla \log \nu_t + K_t \varphi\} \nu_t] \\ &= \frac{1}{2} \int \nabla (|\nabla \phi_t|^2) \cdot \{\nabla \log \nu_t + K_t \varphi\} d\nu_t \\ &= \frac{1}{2} \int \nabla (|\nabla \phi_t|^2) \cdot \nabla \log \nu_t - \Delta |\phi_t|^2 \nu_t - \int \mathcal{L}_t |\phi_t|^2 d\nu_t \\ &= - \int \mathcal{L}_t |\phi_t|^2 d\nu_t \end{aligned}$$

Sketch of proof for the inhomogeneous O.-U.

therefore

$$\begin{aligned}\partial_t \mathbb{E}_{\nu_t} |\nabla \phi_t|^2 &= \mathbb{E}_{\nu_t} \left[\mathcal{L}_t |\nabla \phi_t|^2 - 2 \langle \nabla \mathcal{L}_t \phi, \nabla \phi \rangle \right] \\ &= -2 \mathbb{E}_{\nu_t} \Gamma_{2,t} \phi_t.\end{aligned}$$

So the derivative of the Rayleigh ratio is given by

$$\left(\mathbb{E}_{\nu_t} (\phi_t)^2 \right)^2 R'_\phi(t) = -2 \mathbb{E}_{\nu_t} \Gamma_{2,t} \phi_t \mathbb{E}_{\nu_t} (\phi_t)^2 + \left(\mathbb{E}_{\nu_t} |\nabla \phi_t|^2 \right)^2$$

Sketch of proof for the inhomogeneous O.-U.

Recall that $\Lambda_t = \Delta - \nabla \log(\nu_t) \cdot \nabla$ is an homogeneous generator admitting ν_t as an equilibrium distribution.

$$\begin{aligned}\mathbb{E}_{\nu_t} |\nabla \phi_t|^2 &= -\mathbb{E}_{\nu_t} \phi_t \Lambda_t \phi_t \\ &\leq \sqrt{E_{\nu_t} \phi_t^2 E_{\nu_t} (\Lambda_t \phi_t)^2} \\ &= \sqrt{E_{\nu_t} \phi_t^2 E_{\nu_t} \Gamma_{2,t}^\Lambda \phi_t}.\end{aligned}$$

Hence

$$\left(\mathbb{E}_{\nu_t} (\phi_t)^2 \right) R'_\phi(t) \leq \mathbb{E}_{\nu_t} (\Gamma_{2,t}^\Lambda - 2\Gamma_{2,t}) (\phi_t)$$

Sketch of proof for the inhomogeneous O.-U.

$$\left(\mathbb{E}_{\nu_t} (\phi_t)^2 \right) R'_\phi(t) \leq \mathbb{E}_{\nu_t} (\Gamma_{2,t}^\Lambda - 2\Gamma_{2,t}) (\phi_t)$$

Assumption in that case

$$(\Gamma_{2,t}^\Lambda - 2\Gamma_{2,t}) \leq \dot{k}_t \Gamma_t = \frac{\dot{k}_t}{2} |\nabla \cdot|^2$$

so

$$R'_\phi(t) \leq \frac{\dot{k}_t}{2} \frac{\mathbb{E}_{\nu_t} |\nabla \phi_t|^2}{\mathbb{E}_{\nu_t} (\phi_t)^2} = \frac{\dot{k}_t}{2} R_\phi(t)$$

and the proof is concluded by using Grönwall's lemma.

Mixing time along the Polchinski flow

Thm (S' 24, and work in progress with Y. Shenfeld 24+)

Let $\nu_t(d\varphi) \simeq \gamma_{C_\infty - C_t}(d\varphi) e^{-V_t(\varphi)}$ be the Polchinski flow.
Then the rescaling $\hat{\nu}_t$ by $\dot{C}_t^{1/2}$ is the law of an inhomogeneous O.-U. starting from ν_0 . Moreover, the dynamical condition

$$\frac{d}{dt} \Gamma_t^\Lambda + (\Gamma_{2,t}^\Lambda - 2\Gamma_{2,t}) \leq \dot{k}_t \Gamma_t$$

is equivalent to the Bauerschmidt-Bodineau condition

$$\dot{C}_t \nabla^2 V_t \dot{C}_t - \frac{1}{2} \ddot{C}_t \geq \dot{\lambda}_t \dot{C}_t$$

and $\dot{k}_t, \dot{\lambda}_t$ are linked by the largest eigenvalue of K_t .

→ control of the mixing time along the Polchinski flow of the $\phi_{2,3}^4$ models.

Lipschitz transportation of measures

Lipschitz transportation of measures

A Lipschitz map

$$T : \mathbb{R}^d \rightarrow \mathbb{R}^d$$

such that

$$T^\#(\mu) = \nu$$

transfers good properties of μ to ν .

(e.g. concentration, spectrum, relaxation time).

→ If $d\nu = f \, dx$, then $d\mu = (f \circ T) |\det \nabla T| \, dx$.

→ Goal : start with ν , and try to construct a Lipschitz transport from the Gaussian to ν .

Caffarelli's contraction theorem

Thm (Caffarelli 2000)

If $d\nu(x) = e^{-V(x)}dx$ and $\nabla^2 V \geq I_d$ then the Brenier O.T. map which pushes forward the standard Gaussian to ν is 1-Lipschitz.

Two proofs :

- Caffarelli's proof by analysing the Monge-Ampère PDE

$$\det \nabla^2 \Phi(x) = \frac{\gamma(x)}{e^{-V(\nabla \Phi(x))}}$$

- By entropy-regularized O.T.
(Fathi-Gozlan-Prodhomme (19), Chewi-Pooladian (23))

The Langevin map

Continuity equation for the Ornstein-Uhlenbeck process :

$$\partial_t \nu_t = \nabla \cdot (\xi_t \nu_t), \quad \xi_t = \nabla \log \frac{d\nu_t}{d\gamma}.$$

Gives a family of diffeomorphisms

$$\begin{cases} \partial_t S_t = -\xi_t(S_t) \\ S_0 = I_d \end{cases}$$

Thm (Kim-Milman 10)

The Langevin map S_∞^{-1} pushes γ to ν_0 and

$$\|\nabla S_\infty^{-1}\|_\infty \leq \exp \left\{ - \int_0^\infty \lambda_{\min}(\nabla \xi_t) dt \right\}.$$

The Langevin map

$$\partial_t \nu_t = \nabla \cdot (\xi_t \nu_t), \quad \xi_t = \nabla \log \frac{d\nu_t}{d\gamma}.$$

Thm (Kim-Milman 10)

The Langevin map S_∞^{-1} pushes γ to ν_0 and

$$\|\nabla S_\infty^{-1}\|_\infty \leq \exp \left\{ - \int_0^\infty \lambda_{\min}(\nabla \xi_t) dt \right\}.$$

Diffusion models in ML :

- estimate score function ξ_t from a sample $X_1, \dots, X_n \sim \nu_0$
- generate new samples $Y \sim \nu_0$ from Gaussian samples.

Renormalization Group for the Langevin map

Thm (Shenfeld 22)

The driving vector is linked with the renormalized potential :

$$-\xi_{t/2}(\varphi) = \dot{C}_t^{1/2} \nabla V_t(\dot{C}_t^{1/2} \varphi)$$

when C_t is the heat kernel decomposition.

As a consequence, Bauerschmidt-Bodineau multiscale criterion
 $\Rightarrow$ Langevin map is Lipschitz.

- $\rightarrow$ Lipschitz constant does not depend on the energy cutoff ε
- $\rightarrow$ gives a Lipschitz transport map from the GFF to the continuum sine-Gordon model

The Polchinski map

- Polchinski map for GFF → $\phi_{2,3}^4$
- Result by Bauerschmidt-Bodineau-Dagallier (23)
- Still something to understand for the Lipschitz constant.

Thanks

Thanks for your attention !!

$$\sum_{n=1}^{\infty} n = -\frac{1}{12}$$